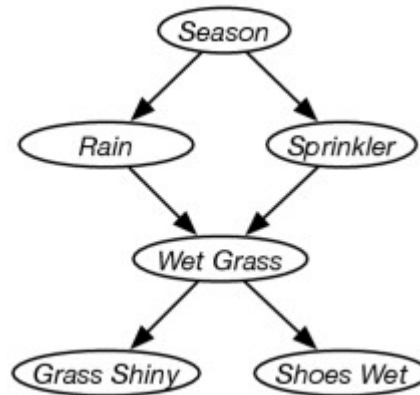


Lecture 4

Answering Causal Questions



Announcements

- ❖ Lab 1 next Wednesday
 - First time getting your hands on a dataset!

Last Time

- ❖ Analyzed the dining hall dataset
 - Discovered a bunch of interesting patterns
 - Formulated hypotheses for further investigation
- ❖ Causal questions
 - Distinguished causal from associational questions

Today

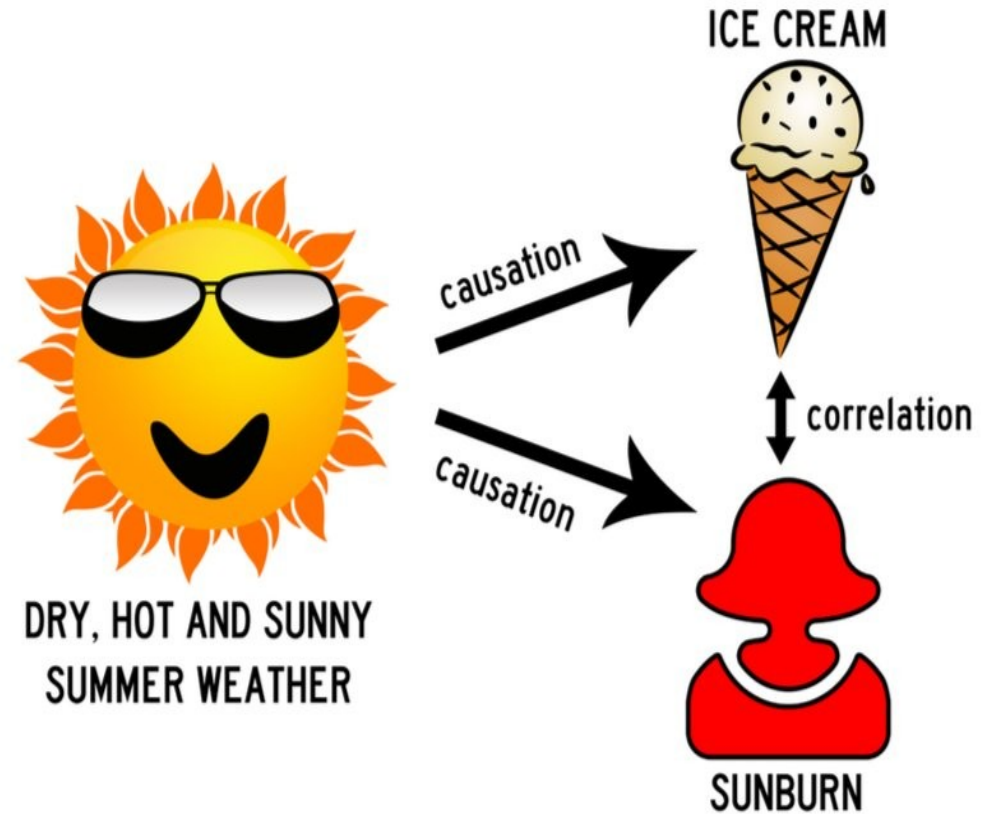
- ❖ Confounding
 - Two associated variables have a common cause
- ❖ Cholera outbreak in 1850s London
 - Interesting case study in causal reasoning
- ❖ Randomization and natural experiments

Confounding

- ❖ Some correlated variables have common causes
- ❖ We call those confounders

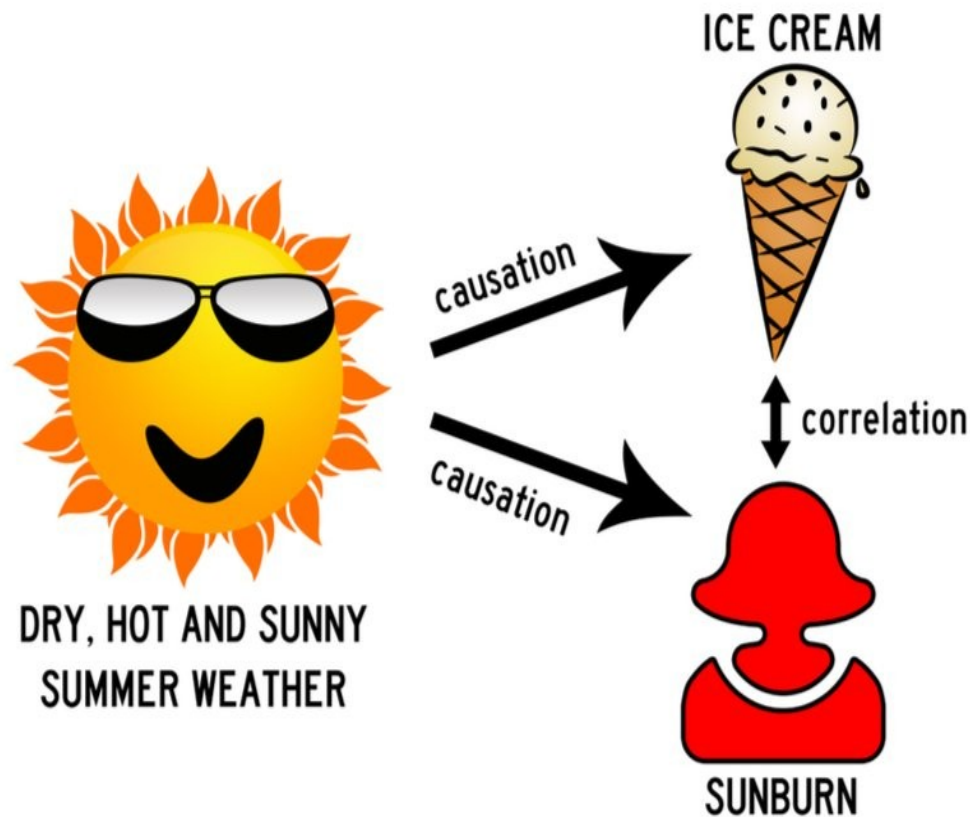
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- ❖ Some correlated variables have common causes
- ❖ We call those confounders
- ❖ Sunny weather causes both ice cream consumption and sunburns



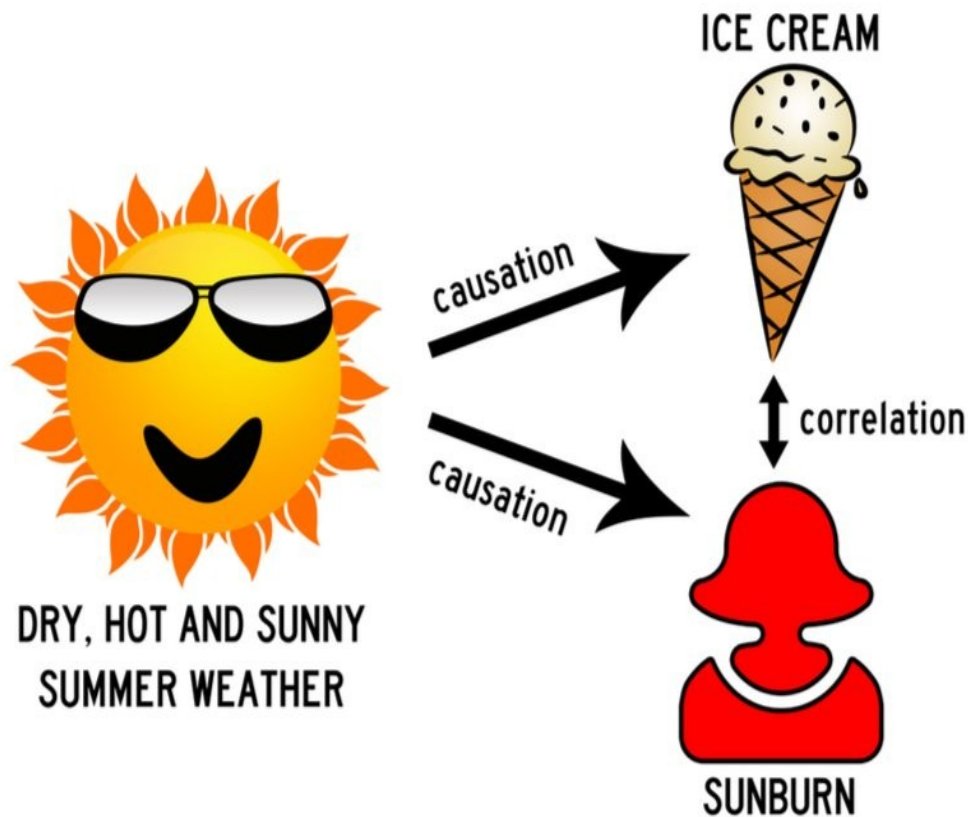
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Confounding

- ❖ Some correlated variables have common causes
- ❖ We call those confounders
- ❖ Sunny weather causes both ice cream consumption and sunburns
- ❖ How pervasive is confounding?
 - Many confounders will not even be observed



Case Study: London 1854

- ❖ Plotted cholera deaths and water pump locations



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- ❖ Observed many homes with deaths used the pump on Broad street
- ❖ Conjecture: Broad street pump causing cholera spread



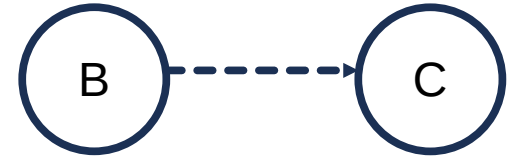
Case Study: London 1854

- ❖ Plotted cholera deaths and water pump locations
- ❖ Observed many homes with deaths used the pump on Broad street
- ❖ Conjecture: Broad street pump causing cholera spread
- ❖ Intervention: removed handle and deaths decreased



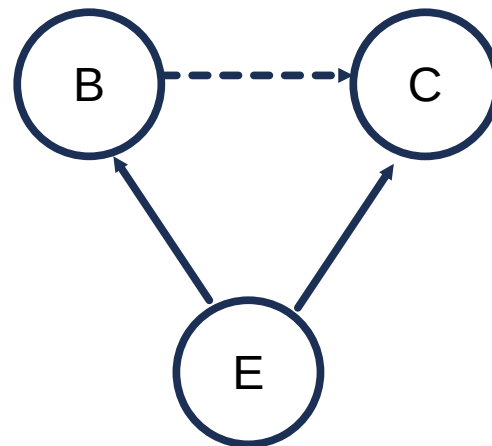
Causal Question

- ❖ Is the evidence sufficient to establish causality?



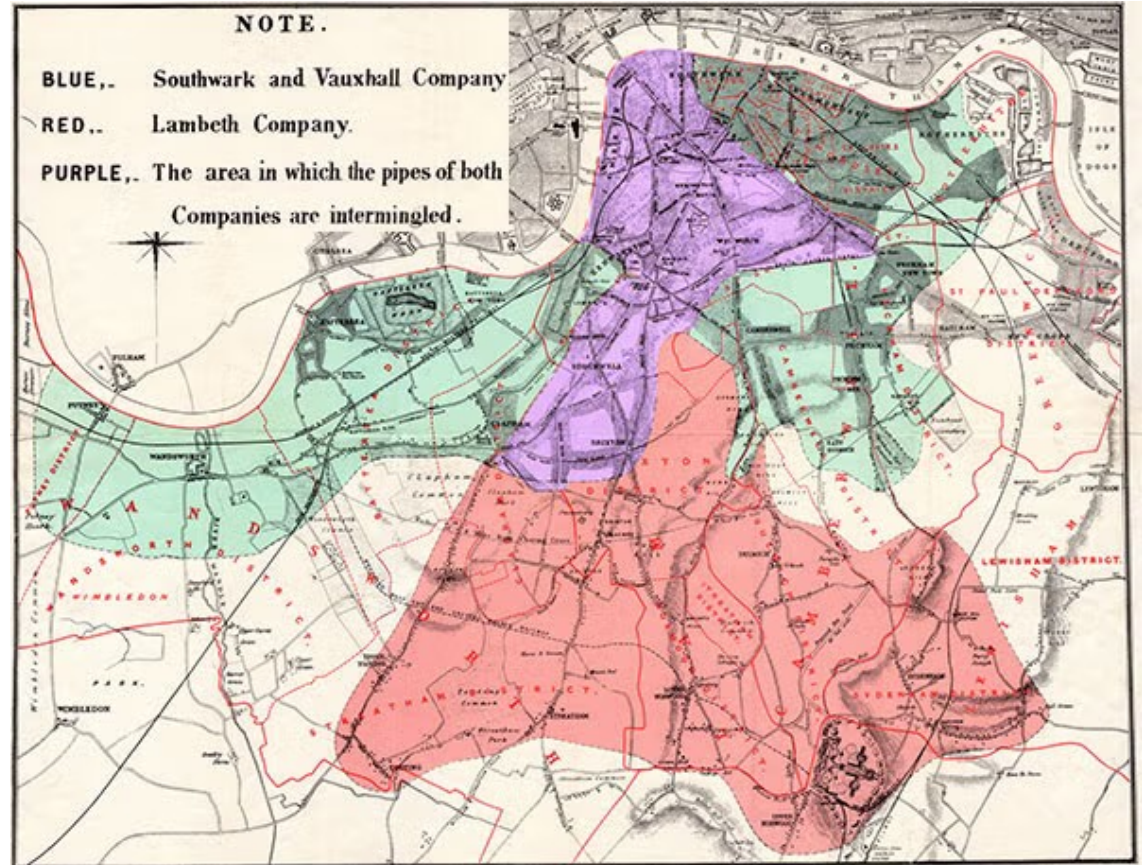
Causal Question

- ❖ Is the evidence sufficient to establish causality?
- ❖ Not necessarily
 - The evolution of the outbreak itself is a confounder since the disease was already tapering off when the pump was removed
- ❖ We cannot separate the effect of removing the pump from the trajectory of the disease



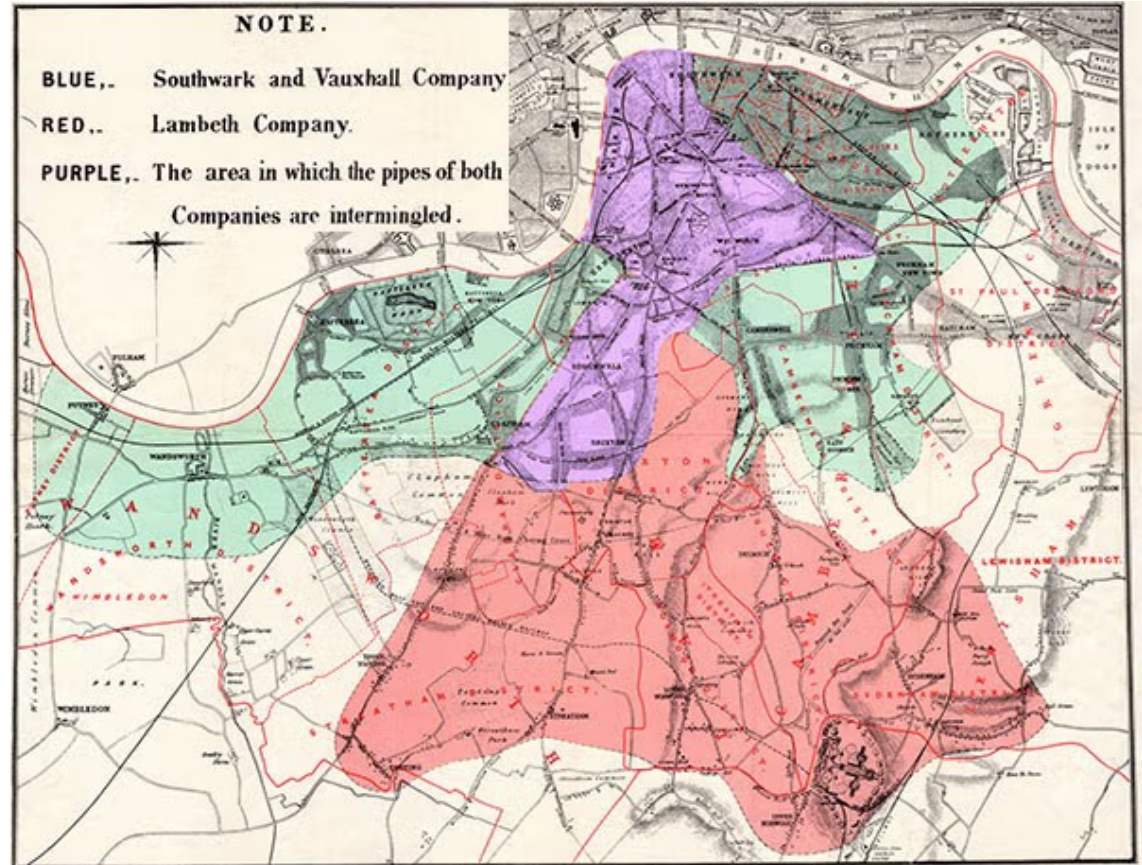
Followup Study

- ❖ Two water companies
- ❖ Lambeth sources water upstream of sewage; S&V downstream
- ❖ Service areas overlap
- ❖ In overlap, S&V cholera rate 9x higher



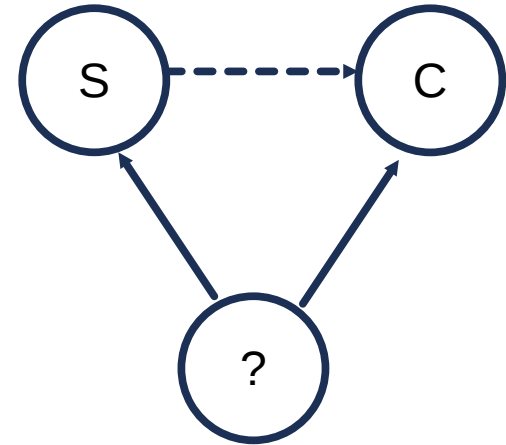
Followup Study

- ❖ Two water companies
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- ❖ Can we finally say that sewage is causing cholera?



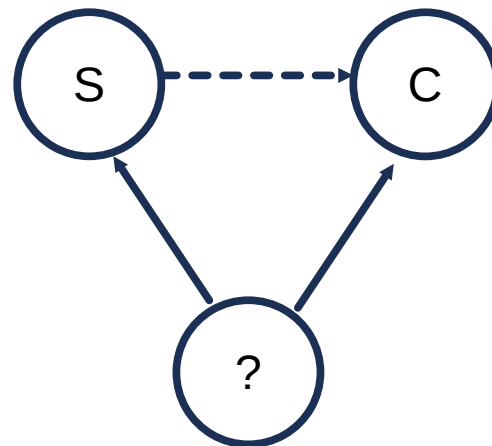
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- ❖ Can you think of any confounders?



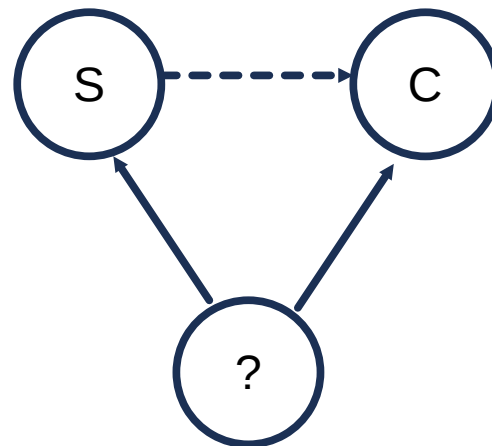
Causal Question

- ❖ Can you think of any confounders?
 - **Possibility:** neighborhood status
 - In the overlap region, it may be that poor neighbors are more likely to both have sewage in their water and die from cholera



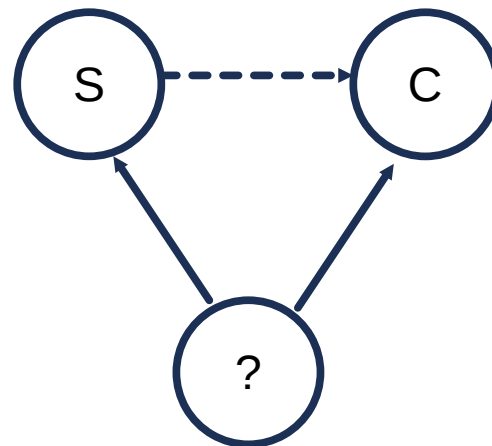
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Randomization

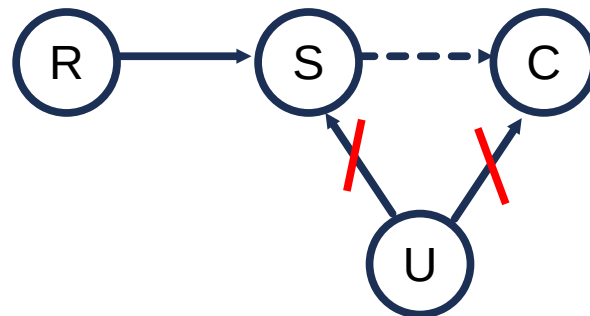
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- ❖ Idea: randomly assign units to treatment or control
 - Why does this solve our issues with confounding?

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 - Why does this solve our issues with confounding?
 - No other variable U jointly causes both treatment and outcome



Types of Studies

- ❖ **Experimental:** researchers randomize treatment and measure outcome
 - Example: randomized control trial (RCT) for a new cancer drug
 - Not suitable for all domains and may run into practical difficulties

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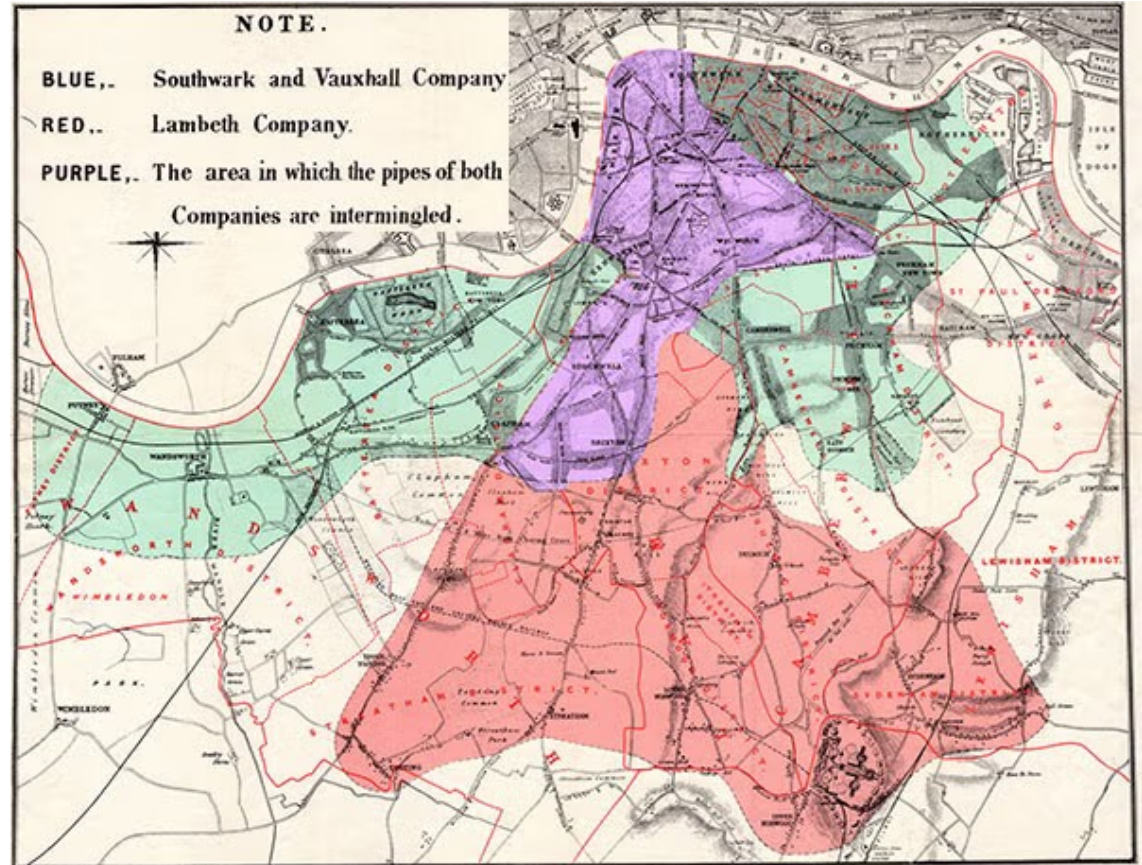
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- ❖ **Observational:** measure treatment and outcome but no intervention
 - Unless we have more info, restricted to measuring correlations
 - Can still be useful!
- ❖ **Natural Experiment:** no intervention but treatment is random-ish
 - Idea: may be something that causes treatment but does not otherwise affect outcome
 - Under some assumptions, we can infer causality

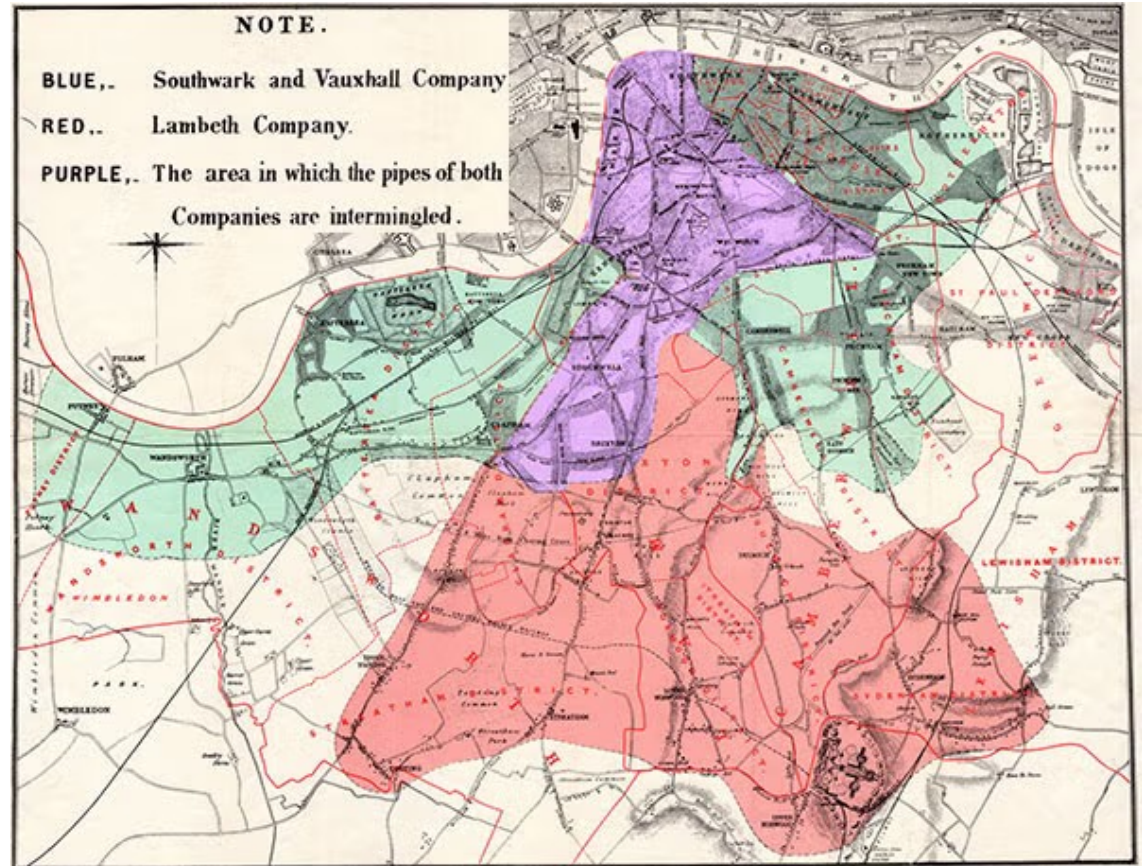
Followup Study

- ❖ In the overlap region, water company was more or less random
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Followup Study

- ❖ In the overlap region, water company was more or less random
- ❖ Lambeth houses and S&V houses otherwise looked alike
- ❖ If the randomization is good enough, we can conclude causality



Next Time

- ❖ Read DS Ch. 1
- ❖ Python & Pandas from scratch